

SELECTED INSTRUCTIONS

1.Instruction:MOV A,#data

Functional Category: Data Transfer

Operands: A, immediate data

Operation performed: loads a 8-bit value data into accumulator.

Registers/Memory Affected: ACC

Flag affected :None

Test Case: MOV A, #05 which will be ACC=05

2.Instruction:MOV Rn ,#data

Functional Category: Data Transfer

Operands: R0-R7, immediate data

Operation performed: loads a 8-bit value data into register Rn.

Registers/Memory Affected: R0-R7

Flag affected :None

Test Case: MOV R1, #03 which will be R1=03

3.Instruction:ADD A,Rn

Functional Category: Arithmetic

Operands: A, R0-R7

Operation performed: adds Rn to A

Registers/Memory Affected: A

Flag affected : carry flag

Test Case: if A=10,R0=5 then A=15

4.Instruction:ADD A,#data

Functional Category: Arithmetic

Operands: A, immediate data

Operation performed: adds value to A

Registers/Memory Affected: A

Flag affected : carry flag

Test Case: if A=20,ADD A,#10 then A=30

5.Instruction:INC A

Functional Category: Arithmetic

Operands: A,

Operation performed: increments A by 1

Registers/Memory Affected: A

Flag affected : NONE

Test Case: if A=10,INC A then A=11

6.Instruction:DEC A

Functional Category: Arithmetic

Operands: A,

Operation performed: Decrements A by 1

Registers/Memory Affected: A

Flag affected : NONE

Test Case: if A=10,DEC A then A=9

7.Instruction:ANL A,Rn

Functional Category: Logical

Operands: A, R0-R7

Operation performed: performs bitwise AND operation between A and Rn

Registers/Memory Affected: A

Flag affected : NONE

Test Case: if A=0Fh,R0=03h ->A=03h

8.Instruction:SJMP add

Functional Category: Branch/control

Operands: Relative address

Operation performed: performs an unconditional short jump

Registers/Memory Affected: PC

Flag affected : NONE

Test Case: if SJMP 0008 then PC will turn 0006 to 0008

9.Instruction:HALT

Functional Category: termination

Operands: none

Operation performed: instruction execution stops

Registers/Memory Affected: none

Flag affected : NONE

Test Case: HALT then instruction execution is Halted.